

Table 1: Distance upper bounds for regular  $m$ -gons

| $m$ | Shor     | RLT      | KRON     | Full SEP |
|-----|----------|----------|----------|----------|
| 3   | 1.414214 | 1.414214 | 1.174750 | 1.114373 |
| 4   | 1.414214 | 1.414214 | 1.133203 | 1.001735 |
| 5   | 1.414214 | 1.414214 | 1.163184 | 1.025778 |
| 6   | 1.414214 | 1.414214 | 1.133203 | 1.001735 |
| 7   | 1.414214 | 1.414214 | 1.149584 | 1.010315 |
| 8   | 1.414214 | 1.414214 | 1.133203 | 1.001735 |
| 10  | 1.414214 | 1.414214 | 1.133203 | 1.001735 |
| 12  | 1.414214 | 1.414214 | 1.133203 | 1.001735 |
| 16  | 1.414214 | 1.414214 | 1.133203 | 1.001735 |

Table 2: Results for normalized random instances

| $m$ | cases | Shor gap<br>median/max | RLT gap<br>median/max | KRON gap<br>median/max | SEP gap<br>median/max |
|-----|-------|------------------------|-----------------------|------------------------|-----------------------|
| 4   | 10    | 0.462/0.525            | 0.454/0.525           | 0.152/0.234            | 0.084/0.219           |
| 6   | 10    | 0.445/0.495            | 0.426/0.494           | 0.203/0.264            | 0.144/0.257           |
| 8   | 10    | 0.480/0.506            | 0.472/0.505           | 0.268/0.355            | 0.217/0.351           |
| 10  | 10    | 0.526/0.618            | 0.526/0.607           | 0.318/0.389            | 0.249/0.332           |
| 15  | 10    | 0.537/0.624            | 0.537/0.624           | 0.358/0.402            | 0.315/0.387           |
| 20  | 10    | 0.581/0.613            | 0.581/0.613           | 0.402/0.430            | 0.365/0.402           |

## Noxious-Facility Computational Results

The equations below are placeholders so this fragment compiles independently. Copy only the block marked “BEGIN PAPER INSERT” into paper/Lorentz.tex.

$$p_i \in \mathbb{R}^2, \quad i = 1, \dots, 4. \quad (1)$$

Table 3: Bounds for instance with points (1)

| method      | distance upper bound |
|-------------|----------------------|
| Shor        | 1.2174074784         |
| RLT         | 1.0382899391         |
| KRON        | 0.8379375081         |
| Full SEP    | 0.7257975088         |
| exact value | 0.7256779914         |